

JEE(Main+Advanced) : Leader & Enthusiast Course
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,D	A,C,D	A,C,D	B,C,D	A,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.40	3.80	5.00	0.04	1.00	9.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	8	8	2	6	6

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,C,D	A,B,C	A,C,D	A,B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.00	1.92	3.33	4.00	9.20	2.98
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	8	9	4	6	5

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C,D	A,B	A,C,D	B,D	A,B,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.00	10.00	600.00	8.75	25.00	9.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	9	3	5	3	1	2

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,D)


 Maximum acceleration of 2kg = $0.2g = 2 \text{ m/s}^2$

Lets assume both are moving together

$$a_{\text{sys}} = \frac{35}{9} > (a_{\text{max}})_{2\text{kg}}$$

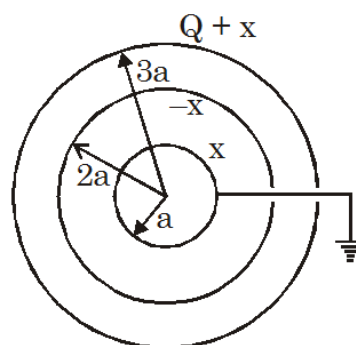
both will move separately

$$f_r = 0.2 \times 2 \times 10 = 4\text{N}$$

$$a_7 = \frac{35 - 4}{7} = \frac{31}{7}$$

$$a_{2\text{kg}} = 2\text{m/s}^2$$

2. Ans (A,C,D)



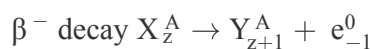
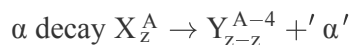
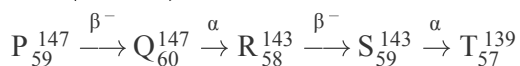
$$\frac{kx}{a} + \frac{k(-x)}{2a} + \frac{k(Q+x)}{3a} = 0$$

$$x - \frac{x}{2} + \frac{Q+x}{3} = 0$$

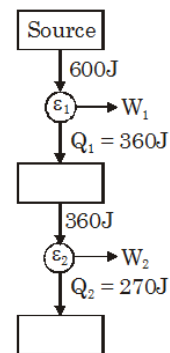
$$6x - 3x + 2Q + 2x = 0$$

$$5x = -2Q \Rightarrow x = \frac{-2Q}{5}$$

3. Ans (A,C,D)



4. Ans (B,C,D)



$$\eta = \frac{W}{Q_{in}}$$

for $\epsilon-1$

$$0.40 = \frac{W_1}{600} \Rightarrow W_1 = 240$$

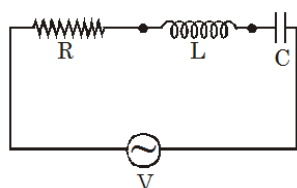
$$Q_1 = 600 - 240 = 360 \text{ J}$$

$$\frac{W_2}{360} = 0.25 \Rightarrow W_2 = 90$$

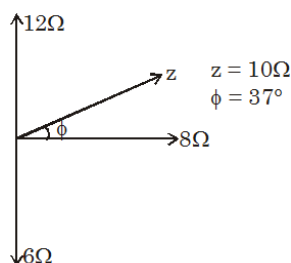
$$Q_2 = 360 - 90 = 270 \text{ J}$$

$$\eta_{\text{system}} = \frac{330}{600} \times 100 = 55\%$$

5. Ans (A,D)



$$V = 50 \sin(\omega t + 37^\circ)$$

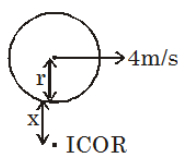
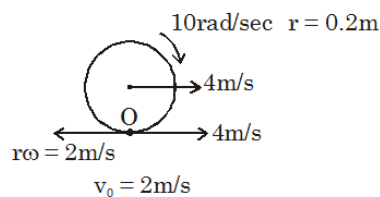


$$i = 5 \sin(\omega t)$$

$$P = \epsilon i = 250 \times \frac{4}{5} = 200 \text{ watt}$$

$$\epsilon_L = (5 \times 12) \sin(\omega t + 90^\circ) = 60 \cos \omega t$$

6. Ans (A,B,C,D)



about ICOR

$$v = r\omega$$

$$4 = (r + x)10$$

$$x = 0.2 \text{ m}$$

$$L_{\text{ICOR}} = I\omega = \left[\frac{mr^2}{2} + m(2r)^2 \right] \omega$$

$$= \frac{9mr^2}{2} \omega = \frac{9 \times 10}{25 \times 2} = 1.8$$

$$L_{\text{CM}} = I_{\text{CM}} \omega = \frac{mr^2}{2} \omega = \frac{1}{25} \times \frac{10}{2} = 0.2$$

SECTION-II

1. Ans (0.40)

Under steady state, no current will flow through capacitor.

$$V_C = 4V$$

$$q = 0.1 \times 4 = 0.4 \mu\text{C}$$

2. Ans (3.80)

$$KE_{\text{max}} = \frac{hc}{\lambda_1} - \phi \quad \lambda_1 = 200 \text{ nm}$$

$$(eV_0 + KE_{\text{max}}) = \frac{hc}{\lambda_{\text{min}}} \quad \lambda_{\text{min}} = 100 \text{ nm}$$

$$10eV + \frac{12400}{2000} - \phi = \frac{12400}{1000}$$

$$10 + 6.2 - \phi = 12.4$$

$$\phi = 16.2 - 12.4 = 3.8 \text{ eV}$$

3. Ans (5.00)

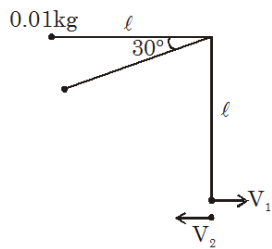
At horizontal level

$$P_0 - \frac{2T}{r_1} + \rho gh_1 = P_0 - \frac{2T}{r_2} + \rho gh_2$$

$$\frac{15 \times 10^{-2}}{10^{-3}} \left(\frac{1}{1} - \frac{2}{3} \right) = 1000 \times 10 \times \Delta n$$

$$\Delta n = \frac{5 \times 10^{-2}}{10} = 5 \text{ mm}$$

4. Ans (0.04)



$$v_1 = \sqrt{2g\ell}$$

$$v_2 = \sqrt{2g(\ell - \ell \sin 30^\circ)}$$

$$\sqrt{2} = 1.414$$

from momentum conservation

$$0.01 v_1 = -0.01 v_2 + 1.205 v$$

$$c = 0.01 (v_1 + v_2) = \frac{0.01 (\sqrt{2} + 1) \sqrt{g\ell}}{1.205} = 0.2$$

for block "1.205 kg"

$$0 = v^2 - 2(\mu g)s$$

$$(0.20)^2 = 2 \mu \times 10 \times \frac{1}{2}$$

$$\mu = 0.004$$

5. Ans (1.00)

$$KE = \int \frac{1}{2} dm v_p^2$$

$$= \int \frac{1}{2} (\mu dx) [10^{-2} \cos(2\pi x - 20\pi t) \times 20\pi]^2$$

$$= \frac{1}{2} \int \mu (400\pi^2 \times 10^{-4}) \cos^2 (2\pi x - 20\pi t) 0$$

$$= \frac{1}{2} \times \frac{1}{2} \times 10^{-2} \times 0.4 \times 1 = 10^{-3} J$$

6. Ans (9.00)

$$f' = \left(\frac{v + v_s}{v - v_s} \right) f_0$$

$$\text{Beat frequency} = f' - f_0 = \frac{2v_s}{v - v_s} f_0$$

$$9 = \frac{2v_s}{v - v_s} f_0$$

$$20 = \frac{4v_s}{v - 2v_s} f_0$$

$$\frac{20}{9} = \frac{2(v - v_s)}{(v - 2v_s)}$$

$$10v - 20v_s = 9v - 9v_s$$

$$11v_s = v = 330 \text{ m/s}$$

$$v_s = 30 \text{ m/s}$$

$$9 = \frac{60}{300} f_0 \Rightarrow f_0 = 45$$

$$c = 9$$

SECTION-III

1. Ans (8)

$$\frac{R}{R_1 + R_2} = \frac{L/4}{3L/4} \dots (1)$$

$$\frac{R + R_1}{R_2} = \frac{2L/3}{L/3} \dots (2)$$

$$3R = R_1 + R_2$$

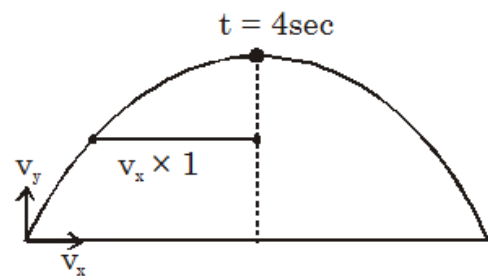
$$R + R_1 = 2R_2$$

$$4R = 3R_2$$

$$\Rightarrow R_2 = \frac{4R}{3}$$

$$R_1 = \frac{5R}{3}$$

2. Ans (8)



$$0 = v_y - 10 \times 4 \Rightarrow v_y = 40 \text{ m/s}$$

$$s_{nth} = u + \frac{a}{2} (2n - 1)$$

$$s_y = 40 - 5(3) = 25$$

$$s_x = v_x \times 1 = v_x$$

$$(v_x^2 + 625)^{1/2} = 5(217)^{1/2}$$

$$v_x^2 = 25 \times 217 - 625 = 25(217 - 25) = 25 \times 192$$

$$v_x = 40\sqrt{3}$$

$$\text{Speed} = \sqrt{v_x^2 + v_y^2} = 80 \text{ m/s}$$

3. Ans (8)

$$80 = 4e\sigma AT_s^3 (40 - T_s)$$

$$ms8 = 4e\sigma AT_s^3 (40 - T_s) \Rightarrow ms = 10$$

$$-ms \frac{dT}{dt} = 4e\sigma AT_s^3 (27 - T_s)$$

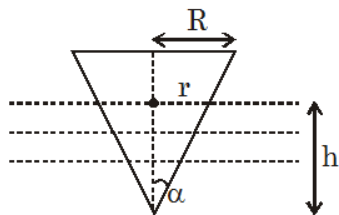
$$10 \times 5 = \frac{80}{(40 - T_s)} (28 - T_s)$$

$$200 - 5T_s = 224 - 8T_s$$

$$3T_s = 24$$

$$T_s = 8$$

4. Ans (2)



$$mg = B$$

$$\rho_{OB} \frac{1}{3} \pi R^2 H g = \rho_L \frac{1}{3} \pi r^2 h g$$

$$\frac{H^3}{h^3} = \frac{\rho_L}{\rho_{OB}} = \frac{64}{27} \Rightarrow \frac{H}{h} = \frac{4}{3}$$

Now it is displaced slightly (x)

$$\rho_L \pi r^2 g x = \rho_{OB} \frac{1}{3} \pi R^2 H a$$

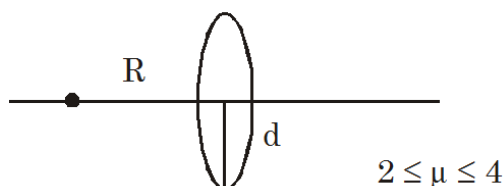
$$a = \frac{3 \rho_L}{\rho_{OB}} \frac{r^2}{R^2 H} g x$$

$$a = 3 \frac{H^3}{h^3} \frac{h^2}{H^2 H} g x = \frac{3 g x}{h}$$

$$a = g x$$

$$T = \frac{2\pi}{\sqrt{g}}$$

5. Ans (6)



$$\frac{1}{f} = \frac{(\mu - 1)}{1} \left(\frac{2}{R} \right)$$

$$f = \frac{R}{2(\mu - 1)}$$

$$f_1 = \frac{R}{2}, f_2 = \frac{R}{6}$$

$$\frac{1}{v_1} + \frac{1}{R} = \frac{1}{R/2} \quad \frac{1}{v_2} + \frac{1}{R} = \frac{6}{R}$$

$$v_1 = R \dots (1) \quad v_2 = \frac{R}{5}$$

$$\Delta v = \frac{4R}{5} = \frac{4}{5} \times 7.5 = 1.5 \times 4 = 6$$

6. Ans (6)

$$(\mu - 1) t = \frac{dy}{D}$$

$$(\mu - 1) = \frac{2 \times 10^{-3} \times 5 \text{ mm}}{0.1 \times .5 \text{ mm}} = 2 \times 10^{-1} = 0.2$$

$$\mu = 1.2$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,B,D)

X = 3s : 2 radial nodes ie, 2 nodal surfaces

Y = 3p_z 1 radial node + 1 angular node : 2 nodal surfaces.

As l increases average distance decreases.

2. Ans (A,B,C,D)

(A) (CH₃)₃SiOH is more acidic due to 2pπ-3dπ back bonding.

(B) (CH₃)N(SiH₃)₂ is less basic due to back bonding.

(C) In NO₂ odd electron is present in hybrid orbital, so it easily undergoes dimerisation.

(D) BCl₃ is better Lewis acid as compared to BF₃ and AlCl₃.

3. Ans (A,B,C)

At constant T,

$$V \uparrow P \downarrow$$

So equilibrium shift towards more no. of moles and on increase of temperature, rxn goes in endo-thermic direction.

5. Ans (A,B,C,D)

(A) I.E. of Li⁺ is higher than He.

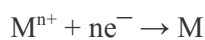
(B) E.G.E. of O < S

(C) Water is strong ligand as compared to chloride.

(D) Ionisation of neutral species is always endothermic.

SECTION-II

1. Ans (4.00)



$$E_{\text{cell}} = 0 - \frac{.06}{n} \log \frac{[M^{n+}]_{\text{Anode}}}{[M^{n+}]_{\text{Cathode}}}$$

$$.03 = - \frac{.06}{n} \log (10^{-2})$$

$$n = 4$$

2. **Ans (1.92)**

dipole of C-A bond $\mu = q \times d$

$$\mu = 1.92 \text{ Debye} = X_{C-A}$$

$$X_{CA_3} = 3X_{CA} \cos(180^\circ - 109^\circ 28')$$

$$= 3X_{CA} \cos 70^\circ 32'$$

$$= 3X_{CA} \times 1/3 = X_{CA}$$

Thus the group moment of CA_3 is same as that of the C-A bond but is directed in the opposite directions so cancel each other

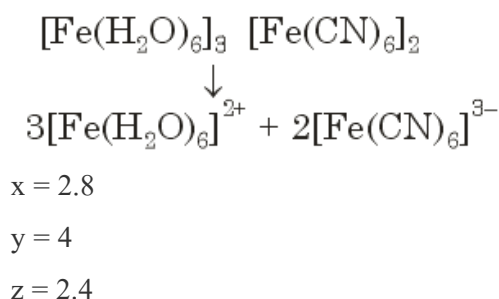
3. **Ans (3.33)**

$$\begin{array}{rcl}
 \text{Li(OH)}_3 & \rightleftharpoons & \text{Li}^{+3} + 3\text{OH}^- \\
 t = 0 & & 0.05 \quad 0.1 \\
 t = \infty & & 0.05 - \frac{0.1}{3} \quad \sim 0 \\
 & & = \left[\frac{0.05}{3} \right] \text{ Initial moles of Li}^{+3} \\
 & & = 0.05 \times 0.1
 \end{array}$$

$$\text{Final moles of Li}^{+3} = \frac{0.05}{3} \times 0.1$$

$$\text{Milli moles precipitated} = 5 - \frac{5}{3} = \frac{10}{3}$$

5. **Ans (9.20)**



6. **Ans (2.98)**

$$\text{Identify value} = \frac{X}{100}$$

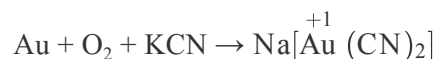
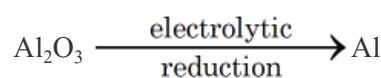
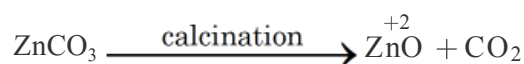
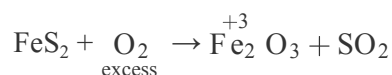
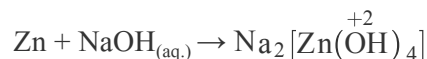
SECTION-III

1. **Ans (6)**

$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = 6.3$$

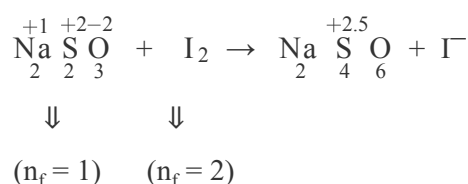
$$\frac{T_2}{T_1} = \frac{P_2 V_2}{P_1 V_1} ; (C_{v,m} = 5/2 \times R)$$

2. **Ans (8)**



$$2 + 3 + 2 + 1 = 8$$

3. **Ans (9)**



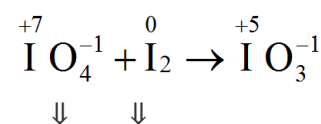
$$\therefore \text{Meq. of Na}_2\text{S}_2\text{O}_3 = \text{meq. of I}_2$$

$$\Rightarrow M \times V_{\text{mL}} \times n_f = \text{mmoles} \times n_f$$

$$\therefore \text{m moles of I}_2 = \frac{0.1 \times 100}{2} = 5$$

$$\therefore \text{m moles of I}_2 = 5 \rightarrow \text{from 1}^{\text{st}} \text{ reaction}$$

Similarly for 2nd reaction



$$n_f = 2 \quad n_f = 10$$

$$\Rightarrow \text{meq. of IO}_4^{-1} = \text{meq. of I}_2$$

$$M \times V_{\text{mL}} \times n_f = (\text{m moles}) \times n_f$$

$$\therefore \text{m moles of I}_2 = \frac{0.2 \times 100 \times 2}{10}$$

$$\therefore \text{m moles of I}_2 = 4 \text{ from 2}^{\text{nd}} \text{ reaction}$$

$$\text{total m moles of I}_2 = 5 + 4 = 9 \text{ m moles}$$

5. Ans (6)

Property	Li	Rest of Group I metals	Mg
Combustion with excess oxygen	O ²⁻ containing anion is main product	(X)	(✓)
Heating with nitrogen gas	(✓)	(X)	Nitride formation
Effect of heat on nitrates	(X)	Brown gas NOT Produces	(X)
Effect of heat on carbonates	(✓)	(X)	Metal oxide with CO ₂
Heating of metal chloride in oxidizing flame	(✓)	Colour on flame	(X)
Physical state of hydrogen carbonates	LiHCO ₃ does not exist as a solid	(X)	(✓)
Solubility of carbonates	Sparingly soluble/insoluble	(✓)	(X)

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,C,D)

We have $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$
 $\Rightarrow \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r} - \frac{1}{r_1} = \frac{1}{2} - \frac{1}{3}$
 $\Rightarrow \frac{r_2 + r_3}{r_2 r_3} = \frac{1}{6} \dots (1)$
 and $r_1 + r_2 + r_3 = 4R + r$
 $\Rightarrow 3 + r_2 + r_3 = 26 + 2$
 $\Rightarrow r_2 + r_3 = 25 \dots (2)$

from (1) and (2), $r_2 = 10$; $r_3 = 15$
 (or vice-versa)

$$\therefore s = \sqrt{\Sigma r_1 r_2} = 15$$

$$\text{and } \frac{r}{r_1} = \frac{s-a}{s} = \frac{2}{3} \Rightarrow a = 5$$

similarly, $b = 13$; $c = 12$ ($a + b + c = 30$)

$$\therefore \Delta = \frac{1}{2} \times 5 \times 12 = 30 \text{ and } 13^2 = 12^2 + 5^2$$

2. Ans (A,B)

$$T_n = \tan^{-1} \left(\frac{2^n}{1 + 2 \cdot (2^n)^2} \right)$$

$$= \tan^{-1} \left(\frac{2^n}{1 + 2^{n+1} \cdot 2^n} \right)$$

$$= \tan^{-1} \left(\frac{2^{n+1} - 2^n}{1 + 2^{n+1} \cdot 2^n} \right)$$

$$T_n = \tan^{-1} 2^{n+1} - \tan^{-1} 2^n$$

$$\therefore \sum_{n=1}^n T_n = \tan^{-1} 2^{n+1} - \tan^{-1} 2$$

$$\Rightarrow f(n) = \tan^{-1} \left(\frac{2 \cdot (2^n - 1)}{1 + 4 \cdot 2^n} \right)$$

$$\lim_{n \rightarrow \infty} f(n) = \frac{\pi}{2} - \tan^{-1} 2 = \cot^{-1} 2$$

3. Ans (A,C,D)

AB will always pass through pole of L.

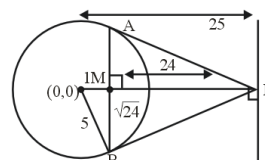
Let pole of L is (x_1, y_1)

$$L : xx_1 + yy_1 = 25$$

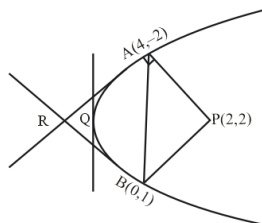
$$\text{also } L : 3x + 4y = 125$$

$$\text{on comparing, } \frac{x_1}{3} = \frac{y_1}{4} = \frac{25}{125}$$

$$\therefore (x_1, y_1) \equiv \left(\frac{3}{5}, \frac{4}{5} \right)$$



4. Ans (B,D)



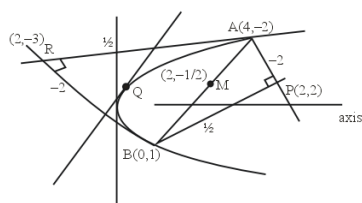
Observe that $AP \perp PB$

$$m_{AP} = -2; m_{PB} = \frac{1}{2}$$

So, ARBP is a rectangle

$\Rightarrow AB$ is a focal chord.

Hence we can find out $R(2,-3)$

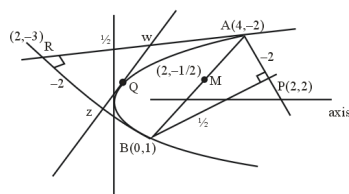


Say $m\left(2, -\frac{1}{2}\right)$ is mid point

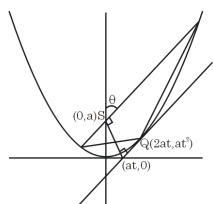
of AB then RMP is parallel to the axis

$\therefore$ Slope of axis is n.d

$\therefore$ Slope of directrix is zero which will pass through $R(2,-3) \Rightarrow y = -3$ is directrix



Since directrix is horizontal so parabola is of the form $x^2 = 4ay$.



AB & tangent at Q are parallel.

So, distance of focus from tangent is $a\sqrt{1+t^2}$

Slope of tangent $= t = m_{AB}$

$$\Rightarrow t = \frac{-3}{4} = \cot \theta$$

$$\text{and } 4a \operatorname{cosec}^2 \theta = 5 \Rightarrow a = \frac{4}{5}$$

$$\therefore \text{Area} = \frac{1}{2} \times AB \times a\sqrt{1+t^2} = \frac{5}{2}$$

5. Ans (A,B,C,D)

$$T_r = \frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$$

$$\sum_{r=1}^i T_r = 1 - \frac{1}{i+1} = \frac{i}{i+1}$$

$$\prod_{i=1}^n \frac{i}{i+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n}{n+1} = \frac{1}{n+1}$$

$$\sum_{n=1}^k \left(\frac{1}{n} \left(\prod_{i=1}^n \left(\sum_{r=1}^i T_r \right) \right) \right) = \sum_{n=1}^k \frac{1}{n(n+1)}$$

$$= \frac{k}{k+1}$$

$$\therefore q - p = 1$$

6. Ans (A,B,C,D)

X = no of dice showing even no.	P = Probability
0	$\left(\frac{3}{6}\right)^5 = \left(\frac{1}{2}\right)^5$
1	${}^5C_1 \cdot \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right)^4 = \frac{5}{2^5}$
2	${}^5C_2 \cdot \left(\frac{1}{2}\right)^2 = \frac{10}{2^5}$
3	${}^5C_3 \cdot \left(\frac{1}{2}\right)^3 = \frac{10}{2^5}$
4	${}^5C_4 \cdot \left(\frac{1}{2}\right)^4 = \frac{5}{2^5}$
5	$\left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)^5$

$$\mu = 1 \cdot \frac{5}{2^5} + 2 \cdot \frac{10}{2^5} + 3 \cdot \frac{10}{2^5} + 4 \cdot \frac{5}{2^5} + 5 \cdot \left(\frac{1}{2}\right)^5$$

$$= \frac{1}{2^5} [5 + 20 + 30 + 20 + 5] = \frac{80}{32} = \frac{5}{2} = 2.5$$

$$\sigma^2 = \frac{1}{2^5} [5 + 40 + 90 + 80 + 25] - \frac{25}{4}$$

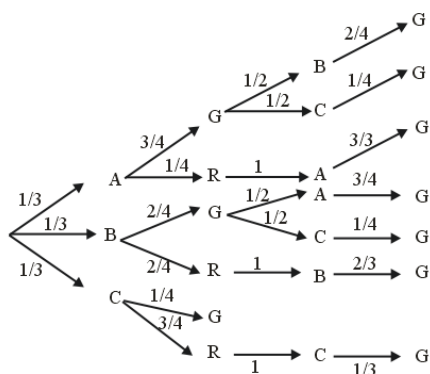
$$= \frac{240}{32} - \frac{25}{4} = \frac{240 - 200}{32} = \frac{40}{32} = \frac{5}{4}$$

$$(D) \frac{1 - \left(\frac{1}{2}\right)^5 - \frac{5}{2^5}}{1 - \left(\frac{1}{2}\right)^5} = \frac{26}{31}$$

SECTION-II

1.

Ans (2.00)



Required

Probability

$$= \frac{1}{3} \left(\frac{3}{4} \cdot \frac{1}{2} \left(\frac{2}{4} + \frac{1}{4} \right) + \frac{1}{4} \cdot 1 \cdot 1 \right)$$

$$+ \frac{1}{3} \left(\frac{2}{4} \cdot \frac{1}{2} \left(\frac{3}{4} + \frac{1}{4} \right) + \frac{2}{4} \cdot 1 \cdot \frac{2}{3} \right) + \frac{1}{3} \left(\frac{1}{4} \cdot \frac{1}{2} \left(\frac{3}{4} + \frac{2}{4} \right) + \frac{3}{4} \cdot 1 \cdot \frac{1}{3} \right)$$

$$= \frac{1}{3} \left[\frac{9}{32} + \frac{1}{4} + \frac{1}{4} + \frac{1}{3} + \frac{5}{32} + \frac{1}{4} \right] = \frac{1}{3} \left[\frac{7}{16} + \frac{13}{12} \right]$$

$$= \frac{1}{3} \left[\frac{21+52}{48} \right] = \frac{73}{3 \times 48} = \frac{m}{n}; \text{ so } \frac{n}{m-1} = \frac{3 \times 48}{72} = 2$$

2. Ans (10.00)

$$PQ = A X B B^T A \text{ [Note } X \cdot X^T = I]$$

$$B \cdot B = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} = 6I$$

$$\therefore PQ = A \cdot X \cdot 6I \cdot X^T A = 6A^2 = 6 \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

$$(PQ)^{10} = \begin{bmatrix} 6^{10} & 0 \\ 0 & 24^{10} \end{bmatrix}$$

$$\therefore \text{tr}((PQ)^{10}) = 6^{10} + 24^{10} = 6^{10} (1 + 4^{10})$$

$$a = 6; b = 4 \Rightarrow a + b = 10$$

3.

Ans (600.00)

$$\text{Say, } A = B + I$$

$$\Rightarrow B = \begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

$$\therefore A^{100} = (B + I)^{100} = 100B + I$$

$$= \begin{bmatrix} 301 & 1 \\ -900 & -299 \end{bmatrix}$$

$$\therefore a_1 - a_4 = 301 - (-299) = 600$$

4.

Ans (8.75)

$$\frac{(k-1)^2}{(k+1)(k+2)} = \frac{k^2 + 3k + 2 - 5k - 1}{(k+1)(k+2)}$$

$$= 1 - \frac{5k+1}{(k+1)(k+2)} = 1 - \frac{5(k+2) - 9}{(k+1)(k+2)}$$

$$= 1 - \frac{5}{k+1} + \frac{9}{(k+1)(k+2)}$$

$$\therefore \sum_{k=0}^{10} \frac{(k-1)^2}{(k+1)(k+2)} {}^{10}C_k = \sum_{k=0}^{10} \left(1 - \frac{5}{k+1} + \frac{9}{(k+1)(k+2)} \right) {}^{10}C_k$$

$$= 2^{10} - \frac{5}{11} \cdot (2^{11} - 1) + \frac{9}{11 \cdot 12} \cdot (2^{10} - 1 - 12)$$

$$= 2^{10} - \frac{10 \cdot 2^{10}}{11} + \frac{5}{11} + \frac{36 \cdot 2^{10}}{11 \cdot 12} - \frac{9}{11 \cdot 12} - \frac{9}{11}$$

$$= \frac{-9 - 4 \times 12}{11 \cdot 12} + \left(\frac{11 \cdot 12 + 36 - 10 \cdot 12}{11 \cdot 12} \right) 2^{10}$$

$$= \frac{4}{11} \cdot 2^{20} - \frac{57}{11 \cdot 12} = \frac{48 \cdot 2^{10} - 57}{132}$$

$$a = 48, b = 57, c = 132, 11 \left(\frac{a+b}{c} \right) = 8.75$$

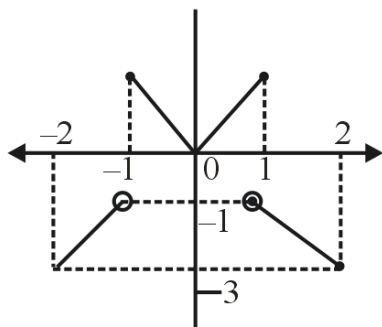
5. Ans (25.00)

$$f(3+x) = f(1-x)$$

$$\Rightarrow f(3+x) = f(x-1)$$

$$\Rightarrow f(x+4) = f(x) \quad \forall x \in \mathbb{R}$$

$\Rightarrow f(x)$ is periodic with period 4 and also even.



$$f(x) = \begin{cases} x & ; 0 \leq x \leq 1 \\ 1 - 2x & ; 1 < x \leq 2 \end{cases}$$

$$f(x) = \begin{cases} |x| & ; 0 \leq x < 1 \\ 1 - 2|x| & ; 1 < x \leq 2 \end{cases}$$

$$\int_0^{100} f(x) dx = 25 \int_{-2}^2 f(x) dx = 50 \int_0^2 f(x) dx$$

$$= 50 \left[\left(\frac{1}{2} \times |x| \right) - \left(\frac{1}{2} (1+3) 1 \right) \right]$$

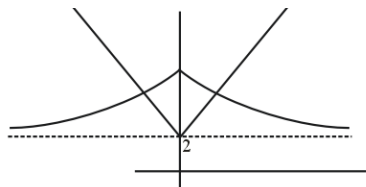
$$I = -\frac{150}{2} = -75$$

$$D = 25 \times 2 = 50$$

$$2D + I = 100 - 75 = 25$$

6. Ans (9.00)

$$f''(x) = \min(e^{-|x|} + 2, |x| + 2)$$



$$f''(x) = 2 \quad \forall x \in (0, \infty)$$

$$f(x) = 2x + a \quad \forall x > 0$$

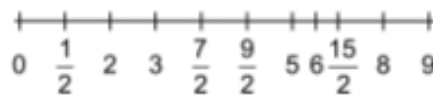
$$f(x) = x^2 + ax + b \quad \forall x > 0$$

$$\therefore L = \lim_{x \rightarrow \infty} \frac{3x^2 - \frac{3}{x^2+1} - 4f'(x)}{f(x)} = 3$$

$$[L^2] = 9$$

SECTION-III

1. Ans (9)



$$f\left(\frac{x+13}{2}\right) = f\left(\frac{3-x}{2}\right)$$

$$f(x) = f(8-x)$$

$$f'(x) = -f'(8-x)$$

$$f'\left(\frac{1}{2}\right) = -f'\left(\frac{15}{2}\right) = 0$$

$$f'(2) = -f'(6) = 0$$

$$f'(3) = -f'(5) = 0$$

$$f'\left(\frac{9}{2}\right) = -f'\left(\frac{7}{2}\right) = 0$$

$$f'(0) = -f'(8); h(x) = \frac{d}{dx}(f'(x) f'''(x))$$

Clearly : $h(x)$ has minimum 18 zeroes

2. Ans (3)

$$\left(1 + e^{\frac{x}{y}}\right) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) dy = 0$$

$$\Rightarrow dx + e^{\frac{x}{y}} dy + y \left(-\frac{x}{y^2}\right) e^{\frac{x}{y}} dy + e^{\frac{x}{y}} dx = 0$$

$$\Rightarrow dx + e^{\frac{x}{y}} dy + y \cdot d\left(e^{\frac{x}{y}}\right) = 0$$

$$\Rightarrow dx + d(y \cdot e^{\frac{x}{y}}) = 0$$

$$\Rightarrow x + y \cdot e^{\frac{x}{y}} = c$$

Passes through (1,1), so $c = 1 + e$

Putting $x = 0$; $y = c = 1 + e$

$$\text{i.e.; } k = 1 + e$$

$$\therefore [k] = [1 + e] = 3$$

3. Ans (5)

$$L_1 : \frac{x+2}{2} = \frac{y+6}{3} = \frac{z-34}{-10}$$

$$\text{and } L_2 : \frac{x+6}{4} = \frac{y-7}{-3} = \frac{z-7}{-2}$$

$$\text{are skew lines as } \begin{vmatrix} -2+6 & -6-7 & 34-7 \\ 2 & 3 & -10 \\ 4 & -3 & -2 \end{vmatrix} \neq 0$$

Direction of line perpendicular to L_1 & L_2

$$\text{is } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -10 \\ 4 & -3 & -2 \end{vmatrix} = (-36\hat{i} - 36\hat{j} - 18\hat{k})$$

$$\text{so } \parallel \text{ vector is } \vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$$

Let line perpendicular to L_1 & L_2 is L

with A and B points respectively on L_1 & L_2

$$A(2\lambda - 2, 3\lambda - 6, -10\lambda + 34)$$

$$B(4\mu - 6, -3\mu + 7, -2\mu + 7)$$

$$AB \text{ is parallel to } 2\hat{i} + 2\hat{j} + \hat{k}$$

$$\therefore \frac{2\lambda - 4\mu + 4}{2} = \frac{3\lambda + 3\mu - 13}{2} = \frac{2\mu - 10\lambda + 27}{1}$$

solving these,

$$2\lambda - 4\mu + 4 = 3\lambda + 3\mu - 13$$

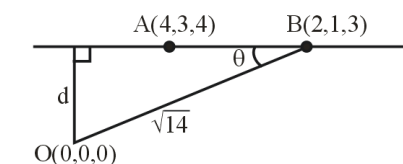
$$\Rightarrow \lambda + 7\mu = 17 \dots(1)$$

$$\text{and } 3\lambda + 3\mu - 13 = 4\mu - 20\lambda + 54$$

$$\Rightarrow 23\lambda - \mu = 67 \dots(2)$$

solving (1) and (2), $\lambda = 3$ and $\mu = 2$

$$\therefore A \equiv (4, 3, 4) \text{ and } B(2, 1, 3)$$



$$\cos \theta = \frac{\vec{AB} \cdot \vec{OB}}{|\vec{AB}| |\vec{OB}|}$$

$$= \frac{4 + 2 + 3}{3 \cdot \sqrt{14}} = \frac{3}{\sqrt{14}}$$

$$d = OB \sin \theta = \sqrt{14} \cdot \frac{\sqrt{5}}{\sqrt{14}} \Rightarrow d = \sqrt{5} \Rightarrow d^2 = 5$$

4. Ans (3)

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \frac{1}{2}$$

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = p + \frac{q}{2} + \frac{r}{2} \dots(1)$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \frac{p}{2} + \frac{q}{2} + r \dots(2)$$

$$0 = \frac{p}{2} + q + \frac{r}{2} \dots(3)$$

$$(1) - (2) \Rightarrow p = r$$

$$\& \text{ from (3) } q = -p$$

$$\therefore \text{ from (2) } \frac{p}{2} + \frac{q}{2} + r = [\vec{a} \ \vec{b} \ \vec{c}] \Rightarrow r = [\vec{a} \ \vec{b} \ \vec{c}]$$

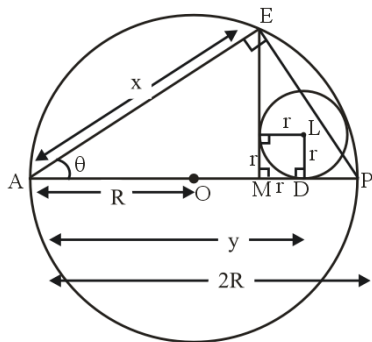
$$\therefore p^2 + q^2 + r^2 = 3[\vec{a} \ \vec{b} \ \vec{c}]^2$$

$$= 3 \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = 3 \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= 3 \left[1 + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} - \frac{1^2}{2^2} - \frac{1^2}{2^2} - \frac{1}{2^2} \right]$$

$$= 3 \left[1 + \frac{1}{4} - \frac{3}{4} \right] = 3 \left[1 - \frac{1}{2} \right] = \frac{3}{2}$$

5. Ans (1)



$$\cos \theta = \frac{x}{2R} = \frac{y-r}{x} \Rightarrow x^2 = 2Ry - 2Rr$$

Also $OL^2 = LD^2 + OD^2$

$$\Rightarrow (R - r)^2 = r^2 + (y - R)^2$$

$$\Rightarrow R^2 + r^2 - 2Rr = r^2 + y^2 - 2yR + R^2$$

$$\Rightarrow 2Ry - 2Rr = y^2$$

$$\therefore x^2 = y^2 \Rightarrow \frac{x}{y} = 1$$

6.

Ans (2)

$$2b = a + c \text{ (Given)}$$

$$\frac{b}{c} \cdot 2 \sin C \cdot \cos C + \frac{c}{b} \cdot 2 \sin B \cdot \cos B$$

$$+\frac{b}{a}2 \sin A. \cos A + \frac{a}{b}.2 \sin B \cos B = 2$$

$$\Rightarrow \frac{2}{2R}b \cos C + \frac{2}{2R}c \cos B + \frac{2b}{2R} \cos A + \frac{2a}{2R} \cos B = 2$$

$$\Rightarrow \frac{2}{2R} [b \cos C + c \cos B + b \cos A + a \cos B] = 2$$

$$\Rightarrow \frac{2}{2R}[a + c] = 2 \Rightarrow \frac{2}{2R}(2b) = 2$$

$$\Rightarrow 4 \sin B = 2 \Rightarrow \operatorname{cosec} B = 2$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

AIOT

12-09-2021

JEE(Main + Advanced) : Leader & Enthusiast Course

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	B	A	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C,D	A,C,D	B,D	B,C,D	A,D	A,B,D	A,B	A,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.75	0.33 to 0.34	0.33 to 0.34	14.83 to 14.84	0.31	15.62		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	A	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B	A,B,C,D	A,B,D	B	C	B	A,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	5.00	12.00	2.00	50.00	391.00 to 392.00	25.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	D	C	B	C,D	A	B,C,D	B,C	B,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.75	0.75	1.40	0.25	1.25	0.25		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

V developed

$$= \frac{1}{2} B \omega l^2 = \frac{1}{2} \times 25 \times 10^{-6} \times \frac{5}{0.25} \times (0.25)^2$$

$$= 15.625 \mu V.$$

$$P = \frac{V^2}{R} = \frac{(16.625 \times 10^{-6})^2}{0.3 \times 10^{-6}} \approx 0.814 mW$$

2. Ans (B)

$$\Delta KE = \frac{1}{2} \mu v_{rel}^2 (1 - e^2)$$

$$\mu = \frac{m \cdot 4m}{5m} = \frac{4}{5} m$$

$$v_{rel} = 3v$$

$$\therefore \Delta KE = \frac{1}{2} \times \frac{4m}{5} 9v^2 (1 - e^2)$$

The energy is utilized for excitation of atoms.

$$\therefore \Delta KE = 51 eV$$

$$\therefore \underbrace{\frac{1}{2} (4m) (4v^2)}_{KE_{He^+}} \frac{9}{10} (1 - e^2) = 51$$

$$\therefore KE_{He^+} = \frac{51 \times 20}{9 (1 - e^2)}$$

$$\therefore \text{For } KE_{He_{min}^+}, e = 0 \Rightarrow KE_{He_{min}^+} = 113.33 eV$$

4. **Ans (A)**

Image formed by the lens will be at a distance of 60 cm on the opposite side of the lens.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-30} = \frac{1}{20} \Rightarrow v = 60 \text{ cm}$$

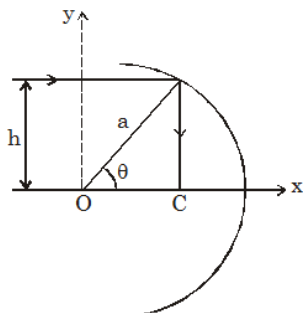
Magnification = -2

$$\therefore d = 0.6 \text{ mm}$$

$$\therefore \beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{0.6 \times 10^{-3}} \text{ m} = 833.33 \mu\text{m}$$

SECTION-I (ii)

5. **Ans (A,C,D)**



We have equation of incoming ray is $y = h$. & equation of mirror surface is $x^2 + y^2 = a^2$.

$\therefore$ equation of reflected ray is

$$y = \frac{h}{a^2 - 2h^2} \left[\left(2\sqrt{a^2 - h^2}x - a^2 \right) \right]$$

The reflected ray crosses x-axis for $y = 0$

$$\therefore x = \frac{a^2}{2\sqrt{a^2 - h^2}}$$

Also $h = a \sin \theta$

$$\therefore y = \frac{a \sin \theta}{a^2 - 2a^2 \sin^2 \theta} \left[2\sqrt{a^2 - a^2 \sin^2 \theta} x - a^2 \right]$$

6. **Ans (A,C,D)**

$$\text{Resistance of the frustum} = \frac{\rho_0 \ell}{\pi r_1 r_2}$$

$$\therefore \text{Rate of heat generated} = \frac{V^2}{R} = \frac{\pi V^2 r_1 r_2}{\rho_0 L}$$

At most temperature rate of heat generation and rate of heat emission becomes equal.

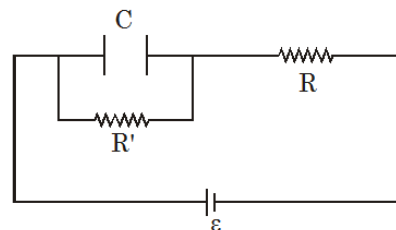
$$\therefore \frac{\pi r_1 r_2 V^2}{\rho_0 \ell} = \sigma \left[\pi r_1^2 + \pi r_2^2 + \pi \ell (r_1 + r_2) \right] \left[T_{\text{max}}^4 - T_0^4 \right]$$

$$\therefore T_{\text{max}} = \left[\frac{r_1 r_2 V^2}{\rho_0 \ell \sigma [r_1^2 + r_2^2 + \ell (r_1 + r_2)]} + T_0^4 \right]^{1/4}$$

7.

Ans (B,D)

We can consider the circuit as



$$\text{Spherical capacitor } C \equiv \frac{4\pi\epsilon_0 k R_1 R_2}{(R_2 - R_1)}$$

$$\text{Spherical resistor } R' = \frac{\rho (R_2 - R_1)}{4\pi (R_1 R_2)}$$

8. **Ans (B,C,D)**

In order to construct a flow curve for the unknown fluid we must measure both the shear rate and the shear stress of the fluid. We are given the cone viscometer with angular velocity ω and torque T . The cone angle, θ is very small, so we make the following approximations $\tan \theta \approx \sin \theta \approx \theta$; $\cos \theta \approx 1$; $R \cos \theta \approx R$

The angular velocity ω should tell us about shear rate. Let us consider a band of fluid at a distance r from the center that is dr wide and h high. (See figure 2) h is given by the geometry of the device and is

$$h = r \tan \theta = r \theta$$

The velocity of the cone would be ωr at the selected radius. The shear rate $\dot{\gamma}$, would then be

$$\dot{\gamma} = \frac{\partial v_\phi}{\partial z} = \frac{\omega r}{r \theta} = \frac{\omega}{\theta}$$

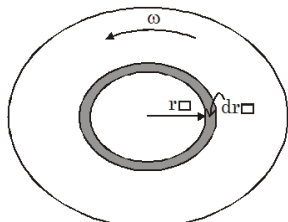
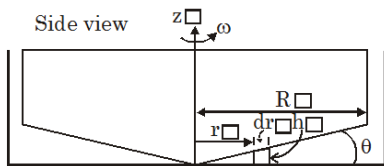
Note that $\dot{\gamma}$ is independent of the radius r .

Next we need to relate the applied torque, T , to the shear stress τ . The shear force acting on the differential surface ring of width dr and radius r would be

$$dF = \tau(r) 2\pi r dr$$

In our case, $\tau(r)$ is actually not a function of r .

$$\tau = \mu \frac{\partial v_\phi}{\partial z} = \frac{\mu \omega}{\theta}$$



The contribution of dF to the torque would then be

$$dT = r dF = 2\pi r^2 dr$$

The total torque would then be

$$T = \int_0^R 2\pi r^2 dr = 2\pi \tau \int_0^R r^2 dr = \frac{2\pi \tau R^3}{3}$$

$$\text{So } \tau = \frac{\mu \omega}{\theta} = \frac{3T}{2\pi R^3} \mu = \frac{3T\theta}{2\pi \times \omega R^3}$$

9. **Ans (A,D)**

We have, $T = mg$

$$\therefore v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{mg}{\mu}}$$

Now since the string is oscillating in 5th harmonic,

$$\therefore \frac{5\lambda}{2} = \ell \Rightarrow \lambda = \frac{2\ell}{5}$$

$$\therefore f = \sqrt{\frac{mg}{\mu}} \times \frac{5}{2\ell}$$

for 'f' to be fundamental frequency.

$$f = \sqrt{\frac{m'g}{\mu}} \cdot \frac{1}{2\ell}$$

$$\therefore m' = 25 \text{ m}$$

Also,

$$f = \sqrt{\frac{mg}{\mu'}} \cdot \frac{1}{2\ell} \Rightarrow \mu' < \mu \text{ (using thin string)}$$

on taking the apparatus to moon,

$$g_{\text{moon}} = \frac{g_{\text{earth}}}{6}$$

$$\therefore \lambda = \frac{1}{f} \sqrt{\frac{mg}{6\mu}} = \left(\frac{1}{f} \sqrt{\frac{mg}{\mu}} \right) \frac{1}{\sqrt{6}}$$

$\therefore$ No standing wave form for given arrangement.

10. **Ans (A,B,D)**

(A) Increasing accelerating potential increase the energy of incident e^- on the target material

thus decreasing the wavelength of emitted x-rays

(B) K_α lines are emitted when there is vacancy in K-shell which means transition from $n = 3$

to $n = 1$ will also be possible.

(C) L-lines are emitted when there is vacancy in $n = 2$, which doesn't ensure vacancy in $n = 1$.

(D) Increasing filament voltage will results in emission of more e^- thus increasing intensity of x-rays.

11. **Ans (A,B)**

Process $A \rightarrow B$

$$W_{AB} = \int P dv = \int \frac{3}{2} T^{1/2} dv = \int \frac{3}{2} T^{1/2} \times \frac{1}{3} RT^{-1/2} dT$$

$$\text{On solving, } W_{AB} = 50 R = 50 \times 8.3 = 415 \text{ J}$$

Process $B \rightarrow C$

$$U = \frac{1}{2} V^{1/2} \frac{3}{2} RT = \frac{1}{2} V^{1/2} \Rightarrow 3PV^{1/2} = 1$$

$$\therefore P = \frac{1}{3\sqrt{V}}$$

$$\text{Now } W_{BC} = \int P dv = \int_{100}^{1600} \frac{1}{3\sqrt{V}} dv = \frac{2}{3} \sqrt{V}$$

$$= \frac{2}{3} [40 - 10] = \frac{2}{3} \times 30 = 20 \text{ J}$$

$$\text{Total } W = 415 + 20 = 435$$

12. **Ans (A,C)**

The total charge is zero, so the dominant term is the dipole. We need the dipole moment of this configuration. It obviously points in the Y direction, and for the southern hemisphere

$\left(\theta : \frac{\pi}{2} \rightarrow \pi \right)$ ρ switches sign but so does Y, so

$$p = \int Y \rho d\tau$$

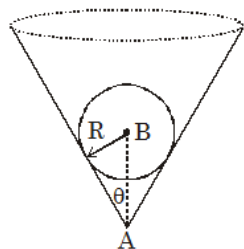
$$= 2\rho_0 \int_{\theta=0}^{\pi/2} r \cos \theta r^2 \sin \theta dr d\theta d\phi$$

$$= 2\rho_0 (2\pi) \int_0^R r^3 dr \int_0^{\pi/2} \cos \theta \sin \theta d\theta$$

$$= 4\pi\rho_0 \frac{R^4}{4} \frac{\sin^2 \theta}{2} \Big|_0^{\pi/2} = \frac{\pi\rho_0 R^4}{2}$$

SECTION-II

1. Ans (2.75)



$$\theta = 37^\circ$$

$$\therefore AB = \frac{R}{\sin \theta} = \frac{5R}{3}$$

$$\text{Also, height of cone } H = 5R \cot \theta = 5R \times \frac{4}{3} = \frac{20R}{3}$$

$$\therefore \Delta U = mg\Delta \left(\frac{5R}{3} \right) + mg\Delta \left(\frac{2H}{3} \right)$$

$$\text{But } \Delta R = R\alpha\Delta T \text{ and } \Delta H = H\alpha\Delta T$$

Putting values,

$$\Delta U = 0.5 \times 10 \times \frac{5}{3} R\alpha\Delta T + mg \frac{2}{3} H\alpha\Delta T$$

$$\frac{25}{3} R \times 10^{-2} + 5 \times \frac{2}{3} \times \frac{20R}{3} \times 10^{-2} = 2.75J$$

2. Ans (0.33 to 0.34)

$$\sigma = -\frac{\frac{dr}{r}}{\frac{d\ell}{\ell}}$$

$$\text{Given } \frac{d\ell}{\ell} = \frac{0.12 \times 10^{-2}}{2} = 0.06 \times 10^{-2}$$

Also, since water is incompressible $\Rightarrow$ volume of water remains same.

$$\therefore A_0 \times 2 = A_N (2 + 0.08 \times 10^{-2})$$

$$\Rightarrow A_N = \frac{A_0 \times 2}{2 + 0.08 \times 10^{-2}}$$

$$\Rightarrow \frac{A_N}{A_0} = \frac{2}{2 + 0.08 \times 10^{-2}}$$

$$\Rightarrow \frac{A_N}{A_0} = \frac{2}{2(1 + 0.04 \times 10^{-2})} = (1 + 0.04 \times 10^{-2})^{-1}$$

$$\Rightarrow \frac{\pi(r_0 + \Delta r)^2}{\pi r_0^2} = 1 - 0.04 \times 10^{-2}$$

$$\Rightarrow \left(1 + \frac{\Delta r}{r_0}\right)^2 = 1 - 0.04 \times 10^{-2}$$

$$\Rightarrow \frac{\Delta r}{r} = -0.02 \times 10^{-2}$$

$$\therefore \sigma = \frac{-\frac{\Delta r}{r}}{\frac{d\ell}{\ell}} = \frac{0.02 \times 10^{-2}}{0.06 \times 10^{-2}} = \frac{1}{3}$$

3. Ans (0.33 to 0.34)

We have angular momentum of chimney

$$L(x) = L_{\text{above}} + L_{\text{below}}$$

$$= \frac{1}{6} \frac{m\omega^2}{L} (L-x)^3 + \frac{1}{3} mx^2 \omega^2$$

$$\text{Since } \tau = \frac{dL}{dt}$$

$$\therefore \frac{d\tau}{dx} = \frac{dL}{dx}$$

$$\text{For maxima, } \frac{d\tau}{dx} = 0$$

$$\therefore x = \frac{L}{3}$$

4. Ans (14.83 to 14.84)

$$\text{Given, } V = 54 \text{ km/hr} = 15 \text{ m/s}$$

$$R = 150 \text{ m, } g = 10 \text{ m/s}^2$$

For meter gauge, distance between rail = 1 m

$$\therefore \text{Angle of bank} = \tan^{-1} \left[\frac{V^2}{Rg} \right] \text{ [No side thrust]}$$

$$\tan^{-1} \left[\frac{15 \times 15}{150 \times 10} \right] = \tan^{-1} \left(\frac{15}{100} \right)$$

$$\therefore h = 1 \sin \theta \text{ [}\theta \text{ is angle of bank]}$$

$$= 14.83 \text{ cm}$$

5. Ans (0.31)

Minimum distance through which the block of 2kg must be displaced so that 5kg block leaves contact is 35 cm.

$\therefore$ The block of 5kg leaves contact when 2kg block is at extreme.

$$\therefore \text{Time taken} = \frac{T}{2} = \frac{2\pi}{2} \sqrt{\frac{2}{200}} = \frac{\pi}{10} \text{ sec}$$

6. Ans (15.62)

Applying work energy theorem at Q & P, we have,

$$\frac{1}{2} mV_Q^2 + mgh_Q = \frac{1}{2} mV_P^2 + mgh_P$$

$$\Rightarrow 0 + mgh_Q = \frac{1}{2} mV_P^2 + mgh_P$$

$$\Rightarrow V_P^2 = 2g(h_Q - h_P) \Rightarrow V_P = \sqrt{2g(h_Q - h_P)}$$

$$= \sqrt{20 \times 3} = \sqrt{62} \text{ m/s}$$

$$\therefore \text{Radial acceleration} = \frac{V^2}{R} = \frac{60}{5} = 12 \text{ m/s}^2$$

$$\text{Tangential acceleration} = 10 \text{ m/s}^2$$

$$\therefore a_{\text{net}} = \sqrt{12^2 + 10^2} = \sqrt{244} \text{ m/s}^2 = 15.62 \text{ m/s}^2$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A)**

$$\Delta \Pi = i(\Delta C) RT$$

$$A = 2 \times (0.1 - 0.05) \times \frac{1}{12} \times 300$$

$$= 0.1 \times \frac{1}{12} \times 300 = \frac{5}{2} = 2.5 \text{ atm}$$

2. **Ans (A)**

ZrSiO₄ - ortho silicate

Be₃Al₂Si₆O₁₈ - cyclic silicate

Al₂(OH)₂[(Si₂O₅)₂] - sheet silicate

3. **Ans (A)**

$$\frac{A_a^0}{A_b^0} = \frac{8}{1} \left| \frac{A_a}{A_b} = \frac{A_a^0 e^{-\lambda_a \times 2}}{A_b^0 e^{-\lambda_b \times 2}} \Rightarrow \frac{1}{1} = \frac{8}{1} e^{(\lambda_b - \lambda_a)2} \right.$$

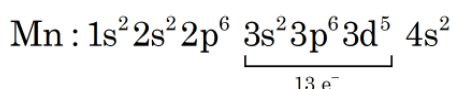
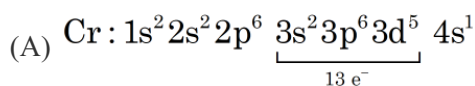
$$\Rightarrow (2)^3 = e^{(\lambda_a - \lambda_b) \times 2 \times 60}$$

$$\Rightarrow 3(\ln 2) = \left(\frac{\ln 2}{20} - \frac{\ln 2}{t_{1/2}} \right) 120$$

$$t_{1/2} = 40 \text{ min.}$$

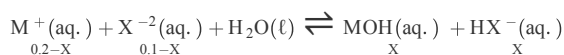
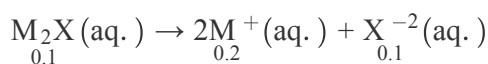
SECTION-I (ii)

5. **Ans (A,B)**



(B) MnO₄⁻ and CrO₄²⁻ both are tetrahedral and coloured

7. **Ans (A,B,D)**



$$K_h = \frac{[MOH][HX^-]}{[M^+][X^{-2}]} = \frac{K_w}{K_{a2}K_b} = \frac{10^{-14}}{10^{-14}}$$

$$\Rightarrow \frac{X^2}{(0.2-X)(0.1-X)} = 1$$

$$\Rightarrow X = \frac{0.2}{3}; [H^+] =$$

8. **Ans (B)**

As per classification of basic radicals

P - AgCl

Q - Bi₂S₃

R - MnS

S - CaCO₃

T - Na⁺

11. **Ans (A,C,D)**

Y - O₃ (Ozone)

X - O₂ (Dioxygen)

3O₂ → 2O₃; ΔH = +ive

SECTION-II

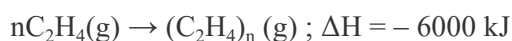
2. **Ans (12.00)**

In [Pt(Br)(NO₂)(NO₃)(SCN)]²⁻, NO₂⁻ and

SCN⁻ are ambidentate ligand. It is [Mabcd]

type square planar complex, so total 12 isomers are possible.

4. **Ans (50.00)**



$$-6000 = (n \times 540) - (2n \times 330) \Rightarrow n = 50$$

5. **Ans (391.00 to 392.00)**

$$\theta_1 = 109^\circ 28'$$

$$\theta_2 = 120^\circ$$

$$\theta_3 = 90^\circ$$

$$\theta_4 = 72^\circ$$

$$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 391^\circ 28'$$

6. **Ans (25.00)**

$$U_{rms} = \sqrt{\frac{3RT}{M}}$$

$$10^5 \times 10^{-2} = \sqrt{\frac{3 \times 25 \times 1000}{3 \times M \times 10^{-3}}}$$

$$M = 25$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

External point must lie on director circle: $x^2 + y^2 = 13$

Let point is $(\sqrt{13} \cos \theta, \sqrt{13} \sin \theta)$

$$\therefore \text{equation of chord of contact } \frac{x\sqrt{13} \cos \theta}{9} + \frac{y\sqrt{13} \sin \theta}{4} = 1$$

If it is tangent to concentric circles: $x^2 + y^2 = a^2$,

$$\Rightarrow \therefore a = \left| \frac{1}{\sqrt{\frac{13 \cos^2 \theta}{81} + \frac{13 \sin^2 \theta}{16}}} \right|$$

$$a_{\max} = \frac{9}{\sqrt{13}}$$

$$a_{\min} = \frac{4}{\sqrt{13}}$$

2. Ans (A)

Solving $y^2 = 5x$ & $x^2 - y^2 = 6 \Rightarrow A \equiv (6, \sqrt{30}), B \equiv (6, -\sqrt{30})$

tangent to 'P' at 'A' $\equiv y(\sqrt{30}) = \frac{5}{2}(x+6) \Rightarrow L \equiv (-6, 0)$

tangent to 'H' at 'A' $\equiv x(6) - y(\sqrt{30}) = 6 \Rightarrow M(1, 0)$

$$\therefore \text{area of } \triangle LAB = \frac{1}{2} (2\sqrt{30}) (12) = 12\sqrt{30}$$

$$\text{area of } \triangle MAB = \frac{1}{2} (2\sqrt{30}) 5 = 5\sqrt{30}$$

$$\text{Ratio} = \frac{12}{5}$$

3. Ans (B)

$$D_1 = 9 - 4a, D_2 = a^2 - 4$$

$$D_1 + D_2 = a^2 - 4a + 5 = (a-2)^2 + 1 > 0$$

4. Ans (A)

$$(a_1)^{1/x_1} = (a_2)^{1/x_2} = (a_3)^{1/x_3} = \dots = (a_n)^{1/x_n} = k, (\text{Let})$$

$$\therefore a_1 = k^{x_1}, a_2 = k^{x_2}, a_3 = k^{x_3} \dots a_n = k^{x_n}$$

Since $a_1, a_2, a_3, \dots, a_n$ are in G.P.

$k^{x_1}, k^{x_2}, k^{x_3} \dots k^{x_n}$ are in G.P.

$\therefore x_1, x_2, x_3, \dots, x_n$ are in A.P.

SECTION-I (ii)

5. Ans (D)

Let A_i = the event that student correctly solve exactly 'i' of the 11 proposed problems. ($i=0,1,2,\dots,11$)

$$\therefore P(A) = \sum_{i=0}^{11} P(A_i)$$

Since student knows to solve exactly half the problem:

$$\therefore P(A_0) = P(A_{11}), P(A_1) = P(A_{10}) \dots P(A_5) = P(A_6)$$

$$\therefore \sum_{i=0}^{11} P(A_i) = 1$$

$$\therefore \sum_{i=0}^5 P(A_i) + \sum_{i=6}^{11} P(A_i) = 1$$

$$\therefore \sum_{i=0}^{11} P(A_i) = P = \frac{1}{2}$$

6. Ans (C)

$$z^{12} - z + (z^{10} + z^9 + \dots + z^2 + z + 1) = 0$$

$$z(z^{11} - 1) + \frac{z^{11} - 1}{z - 1} = 0$$

$$\Rightarrow z^{11} - 1 = 0 \text{ has roots } 1, z_1, z_2, z_3 \dots z_{10}$$

$$\therefore (1)^{2021} + (z_1)^{2021} + (z_2)^{2021} + \dots + (z_{10})^{2021} = 0$$

$$\therefore (z_1)^{2021} + (z_2)^{2021} + \dots + (z_{10})^{2021} = -1$$

7.

Ans (B)

$$x^n - 1 = (x-1)(x-w)(x-w^2) \dots (x-w^{n-1})$$

$$1 + x + x^2 + \dots + x^{n-1} = (x-w)(x-w^2)(x-w^3) \dots (x-w^{n-1})$$

differentiate LHS and RHS both sides

$$1 + 2x + 3x^2 + \dots + (n-1)x^{n-2} = [(x-w^3)(x-w^4) \dots (x-w^{n-1})](2x - (w + w^2)) +$$

$$(x-w)(x-w^2) \frac{d}{dx} [(x-w^3) \dots (x-w^{n-1})]$$

put $x = w$

$$\frac{1 + 2w + 3w^2 + \dots + (n-1)w^{n-2}}{(w-w^2)}$$

$$= (w-w^3)(w-w^4) \dots (w-w^{n-1})$$

8. Ans (C,D)

No. of basic colours = 5

No. of colours formed by mixing

$$= {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 26$$

If n is even

$$\text{No. of ways: } 5^{n/2} \cdot (26)^{n/2} + 5^{n/2} \cdot (26)^{n/2} = 2(130)^{n/2}$$

If n is odd

$$\begin{aligned} \text{No. of ways: } & 5^{\frac{n+1}{2}} \cdot (26)^{\frac{n+1}{2}} + 5^{\frac{n-1}{2}} \cdot (26)^{\frac{n-1}{2}} \\ & = 31(130)^{\frac{n-1}{2}} \end{aligned}$$

9. Ans (A)

$$A = \begin{bmatrix} 5 & 0 & 0 \dots 0 \\ 8 & 0 & 0 \dots 0 \\ 11 & 0 & 0 \dots 0 \\ \vdots & & \\ (3n+2) & 0 & 0 \dots 0 \end{bmatrix} \begin{bmatrix} 5^{-1} & 5^{-2} & 5^{-n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 0 & 0 \dots 0 \\ 8 & 0 & 0 \dots 0 \\ 11 & 0 & 0 \dots 0 \\ \vdots & & \\ 3n+2 & 0 & 0 \dots 0 \end{bmatrix} \begin{bmatrix} 5^{-1} & 5^{-2} & 5^{-n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \dots 0 \\ 8 & 0 & 0 \dots 0 \\ \vdots & \vdots & \vdots \\ 3n+2 & 0 & 0 \dots 0 \end{bmatrix} \begin{bmatrix} 5^{-1} & 5^{-2} & 5^{-n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 0 & 0 \dots 0 \\ 8 & 0 & 0 \dots 0 \\ \vdots & & \\ 3n+2 & 0 & 0 \dots 0 \end{bmatrix} \begin{bmatrix} \Sigma(3i+2)5^{-i} & 0 & 0 \dots 0 \\ 0 & 0 & 0 \dots 0 \\ 0 & 0 & 0 \dots 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5^{-1} & 5^{-2} & 5^{-n} \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^2 = \Sigma(3i+2)5^{-i} A$$

$$\therefore A^n = (\Sigma(3i+2)5^{-i})^{n-1} A$$

$$\text{trace } A^n = (\Sigma(3i+2)5^{-i})^{n-1} (\Sigma(3i+2)5^{-i})$$

$$(\text{trace } A^n)^{1/n} = \Sigma(3i+2)5^{-i}$$

$$= 3 \left(\frac{1}{5} + \frac{2}{5^2} + \dots \infty \right) + 2 \left(\frac{1}{5} + \frac{1}{5^2} + \dots \infty \right) = \frac{23}{16}$$

10. Ans (B,C,D)

$$\begin{aligned} y &= \sum_{r=0}^{n-2} \frac{(r+2)^2 - (r+1)(n-r-1)}{(n-r-1) \cdot {}^nC_{r+1}} \\ &= \sum_{r=0}^{n-2} \left[\frac{r+2}{\frac{n-r-1}{r+2} \cdot {}^nC_{r+1}} - \frac{r+1}{{}^nC_{r+1}} \right] \\ &= \sum_{r=0}^{n-2} \left[\frac{r+2}{{}^nC_{r+2}} - \frac{r+1}{{}^nC_{r+1}} \right] \\ &= n - \frac{1}{n} \end{aligned}$$

11. Ans (B,C)

$$\text{Let } \Delta = \begin{vmatrix} \sin A & \sin B & \sin C \\ \sin(A+kx) & \sin(B+kx) & \sin(C+kx) \\ \cos(A+kx) & \cos(B+kx) & \cos(C+kx) \end{vmatrix}$$

using $R_2 \rightarrow R_2 - (\cos kx)R_1$, $R_3 \rightarrow R_3 + (\sin kx)R_1$,

$$\Delta = \begin{vmatrix} \sin A & \sin B & \sin C \\ \cos A \sin kx & \cos B \sin kx & \cos C \sin kx \\ \cos A \cos kx & \cos B \cos kx & \cos C \cos kx \end{vmatrix} = 0$$

$$\Delta = 0$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \Delta = 0$$

12. Ans (B,D)

$$\text{Let } G(x) = \left(\int_0^x f(t) dt \right)^{2022} \dots (1)$$

$$G'(x) = 2022 f(x) \left(\int_0^x f(t) dt \right)^{2021}$$

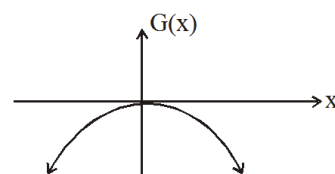
$$\Rightarrow G'(x) = 2022 g(x) \Rightarrow G'(0) = 2022 g(0)$$

also $g(x)$ is decreasing $\Rightarrow g(x) \geq 0 \forall x \in (-\infty, 0)$

and $g(x) \leq 0 \forall x \in [0, \infty)$

$\therefore G'(x) \geq 0$ for $x \leq 0$ and $G'(x) \leq 0$ for $x \geq 0$,

$$G(0) = 0$$



$$\therefore G(x) \leq 0 \forall x \in \mathbb{R}$$

but $G(x) \geq 0 \forall x \in \mathbb{R}$ (from 1)

$$\therefore \Rightarrow G(x) = 0 \Rightarrow \int_0^x f(t) dt = 0 \forall x \in \mathbb{R}$$

$$\therefore f(x) = 0 \forall x \in \mathbb{R}$$

SECTION-II

1. Ans (0.75)

$$\begin{aligned} & \int (4\cot^3 x + \cot^2 x + \cot x - 2) e^x dx \\ \Rightarrow & \int e^x (4\cot x \sec^2 x + \cot^2 x - 3\cot x - 2) dx \\ \Rightarrow & -2e^x \cot^2 x + \int e^x (3\cot^2 x - 3\cot x - 2) dx \\ \Rightarrow & -2e^x \cot^2 x + \int e^x (3\sec^2 x - 3\cot x - 5) dx \\ \Rightarrow & -2e^x \cot^2 x - 3e^x \cot x - 5e^x + c \\ \Rightarrow & -e^x(2\cot^2 x + 3\cot x + 5) + c \\ \therefore & a = 2, b = 3 \end{aligned}$$

2. Ans (0.75)

$$\begin{aligned} & \int_0^1 f(x) dx + \int_0^1 f(x+1) dx + \int_0^1 f(x+2) dx + \dots + \int_0^1 f(x+2020) dx \\ \therefore & \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx + \dots + \int_{2020}^{2021} f(x) dx \\ \Rightarrow & \int_0^{2021} f(x) dx = \frac{3}{4} \end{aligned}$$

3. Ans (1.40)

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{5}{n} \sum_{i=0}^n \left(\frac{7}{5}\right)^{i/n} &= 5 \int_0^1 \left(\frac{7}{5}\right)^x dx \\ &= \frac{2}{\ln 7 - \ln 5} \end{aligned}$$

4. Ans (0.25)

$$\begin{aligned} \bar{b} &= \frac{-\bar{a}}{2} + \frac{\sqrt{3}}{2} \hat{k}; \quad \bar{b} = \frac{-1}{2\sqrt{2}} \hat{i} - \frac{1}{2\sqrt{2}} \hat{j} + \frac{\sqrt{3}}{2} \hat{k} \\ \text{If } |\bar{b} \times \bar{c}| &= |\bar{a} \times \bar{c}| \Rightarrow \text{then } \bar{c} \text{ is perpendicular} \\ &\text{to both } \bar{a} \text{ and } \bar{b} \\ \bar{c} &= \lambda (\bar{a} \times \bar{b}) \quad \& \quad |\bar{c}| = 1 \\ \therefore \bar{c} &= \pm \frac{(\hat{i} - \hat{j})}{\sqrt{2}} \end{aligned}$$

5. Ans (1.25)

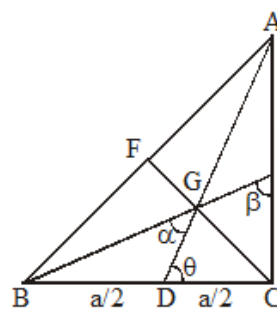
Any point on the line of intersection is $(\lambda + 2, -1, -\lambda)$
For maximum distance
 $= ((\lambda + 2 - 3)\hat{i} + (-1 + 4)\hat{j} + (-\lambda - 5)\hat{k}) \cdot (\hat{i} - \hat{k}) = 0$
 $\Rightarrow \lambda = -2$
Normal $\vec{n} = 3\hat{i} + 5\hat{j} + 3\hat{k}$
equation of plane will be $3x + 5y + 3z = 1$

6. Ans (0.25)

AD: $y = x + 3$
BE: $y = 2x + 4$
 $\Rightarrow G = (-1, 2)$

acute angle ' α ' between the medians is

$$\tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2} \Rightarrow \tan \alpha = \frac{2 - 1}{1 + 2} \Rightarrow \tan \alpha = \frac{1}{3}$$



$$\text{Now } 180 - \alpha + 90^\circ + \theta + \beta = 360^\circ \Rightarrow \alpha = \theta + \beta - 90^\circ$$

$$\begin{aligned} \therefore \cot \alpha &= -\tan(\alpha + \beta) \\ -3 &= \frac{\tan \theta + \tan \beta}{1 - \tan \theta \tan \beta} \Rightarrow -3 = \frac{2b/a + 2a/b}{1 - \frac{2b}{a} \cdot \frac{2a}{b}} \\ \Rightarrow \frac{1}{2} &= 400 \\ \therefore 400 &= 1600 K \\ K &= \frac{1}{4} = 0.25 \end{aligned}$$